

Differentiate Automatically

Automatic Differentiation (Part II)

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Much of this material is based on this blog post:

<https://rufflewind.com/2016-12-30/reverse-mode-automatic-differentiation>

What is Automatic Differentiation (AD)?

To solve optimisation problems using gradient methods we need to compute the gradients (derivatives) of the objective with respect to the parameters.

- In neural nets we're talking about the gradients of the loss function, $\mathcal{L}$ with respect to the parameters θ : $\nabla_{\theta}\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\theta}$
- AD is important - it's been suggested that "Differentiable programming" could be the term that ultimately replaces deep learning¹.

¹<http://forums.fast.ai/t/differentiable-programming-is-this-why-we-switched-to-pytorch/9589/5>

What is Automatic Differentiation (AD)?

Computing Derivatives

There are three ways to compute derivatives:

- Symbolically differentiate the function with respect to its parameters
 - by hand
 - using a CAS
- Make estimates using finite differences
- Use Automatic Differentiation

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Problems

Static - can't “differentiate algorithms”

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Numerical errors - will compound in deep nets

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- **For every variable we wish to compute the gradient with respect to, we have to run the *complete program* again.**
- This is obviously going to be a problem if we're talking about the gradients of a function with very many parameters (e.g. a deep network).
- A solution is **Reverse Mode Automatic Differentiation**.

Reversing the Chain Rule

The chain rule is symmetric — this means we can turn the derivatives upside-down:

$$\frac{\partial s}{\partial u} = \sum_i^N \frac{\partial w_i}{\partial u} \frac{\partial s}{\partial w_i} = \frac{\partial w_1}{\partial u} \frac{\partial s}{\partial w_1} + \frac{\partial w_2}{\partial u} \frac{\partial s}{\partial w_2} + \cdots + \frac{\partial w_N}{\partial u} \frac{\partial s}{\partial w_N}$$

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In doing so, we have inverted the input-output role of the variables: u is some input variable, the w_i 's are the output variables that depend on u . s is the yet-to-be-given variable.

In this form, the chain rule can be applied repeatedly to every input variable u (akin to how in forward mode we repeatedly applied it to every w). Therefore, given some s we expect this form of the rule to give us a program to compute both $\partial s / \partial x$ and $\partial s / \partial y$ in one go. . .

Reversing the chain rule: Example

$$\frac{\partial s}{\partial u} = \sum_i^N \frac{\partial w_i}{\partial u} \frac{\partial s}{\partial w_i}$$

$$x = ?$$

$$y = ?$$

$$a = x y$$

$$b = \sin(x)$$

$$z = a + b$$

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$$\frac{\partial s}{\partial z} = ?$$

$$\frac{\partial s}{\partial b} = \frac{\partial z}{\partial b} \frac{\partial s}{\partial z} = \frac{\partial s}{\partial z}$$

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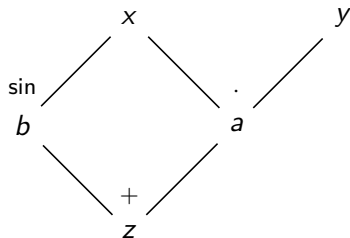
$$\frac{\partial s}{\partial y} = \frac{\partial a}{\partial y} \frac{\partial s}{\partial a} = x \frac{\partial s}{\partial a}$$

$$\begin{aligned} \frac{\partial s}{\partial x} &= \frac{\partial a}{\partial x} \frac{\partial s}{\partial a} + \frac{\partial b}{\partial x} \frac{\partial s}{\partial b} \\ &= y \frac{\partial s}{\partial a} + \cos(x) \frac{\partial s}{\partial b} \\ &= (y + \cos(x)) \frac{\partial s}{\partial z} \end{aligned}$$

Visualising dependencies

Differentiating in reverse can be quite mind-bending: instead of asking what input variables an output depends on, we have to ask what output variables a given input variable can affect.

We can see this visually by drawing a dependency graph of the expression:



Translating to code

Let's now translate our derivatives into code. As before we replace the derivatives $(\partial s / \partial z, \partial s / \partial b, \dots)$ with variables $(gz, gb, \dots)$ which we call *adjoint variables*:

```
gz = ?  
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gy = x * ga  
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If we go back to the equations and substitute $s = z$ we would obtain the gradient in the last two equations. In the above program, this is equivalent to setting $gz = 1$.

This means to get the both gradients $\partial z/\partial x$ and $\partial z/\partial y$ we only need to run the program once!

Limitations of Reverse Mode AD

- If we have multiple output variables, we'd have to run the program for each one (with different seeds on the output variables)². For example:

$$\begin{cases} z = 2x + \sin x \\ v = 4x + \cos x \end{cases}$$

- We can't just interleave the derivative calculations (since they all appear to be in reverse)... How can we make this automatic?

²there are ways to avoid this limitation...

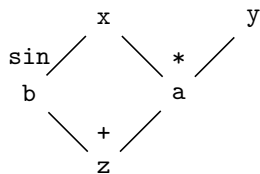
Implementing Reverse Mode AD

There are two ways to implement Reverse AD:

- ① We can parse the original program and generate the *adjoint* program that calculates the derivatives.
 - Potentially hard to do.
 - Static, so can only be used to differentiate algorithms that have parameters predefined.
 - But, efficient (lots of opportunities for optimisation)
- ② We can make a *dynamic* implementation by constructing a graph that represents the original expression as the program runs.

Constructing an expression graph

The goal is to get something akin to the graph we saw earlier:



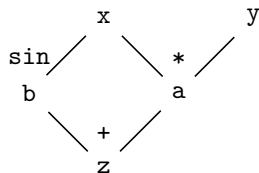
Constructing an expression graph

The “roots” of the graph are the independent variables x and y . Constructing these nodes is as simple as creating an object:

```
class Var:
    def __init__(self, value):
        self.value = value
        self.children = []
        ...
    ...
```

```
x = Var(0.5)
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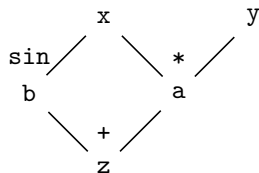
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Each `Var` node can have *children* which are the nodes that depend directly on that node. The children allow nodes to link together in a **Directed Acyclic Graph**.

Building expressions

Each new expression u registers itself as a child of each dependency w_i , together with the local derivative $\partial w_i / \partial u$ used during the reverse sweep:

```
class Var:
    ...
    def __mul__(self, other):
        z = Var(self.value * other.value)

        # weight = dz/dself = other.value
        self.children.append((other.value, z))

        # weight = dz/dother = self.value
        other.children.append((self.value, z))
        return z
    ...
    ...
# "a" is a new Var that is a child of both x and y
a = x * y
```


Computing gradients

Finally, to get the gradients we need to propagate the derivatives. To avoid unnecessarily traversing the tree multiple times we will *cache* the derivative of a node in an attribute `grad_value`:

```
class Var:
    def __init__(self):
        ...
        self.grad_value = None

    def grad(self):
        if self.grad_value is None:
            # calculate derivative using chain rule
            self.grad_value = sum(weight * var.grad() for weight,
                                   var in self.children)
        return self.grad_value
    ...

...
a.grad_value = 1.0
print("da/dx_{}={}".format(x.grad()))
```

Aside: Optimising Reverse Mode AD

- The Reverse AD approach we've outlined is not very space efficient. One way to get around this is to avoid storing the children directly and instead store indices in an auxiliary data structure called a *Wengert list* or *tape*.

³Andreas Griewank (1992) Achieving logarithmic growth of temporal and spatial complexity in reverse automatic differentiation, Optimization Methods and Software, 1:1, 35-54, DOI: 10.1080/10556789208805505

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- Another interesting approach to memory reduction is trade-off computation for memory of the caches. The Count-Trailing-Zeros (CTZ) approach does just this³.
- **But**, in reality memory is relatively cheap if managed well...

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- It allows easy “turning off” of gradient computations through `requires_grad`.
- In-place operations which invalidate data needed to compute derivatives will cause runtime errors, as will variable aliasing...

Reverse mode computes a VJP

For a function $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$, its Jacobian has shape $m \times n$.

Give the output a seed vector $\mathbf{v} \in \mathbb{R}^m$. A reverse sweep computes

$$\underbrace{\bar{\mathbf{x}}^\top}_{1 \times n} = \underbrace{\mathbf{v}^\top}_{1 \times m} \underbrace{\mathbf{J}_f(\mathbf{x})}_{m \times n} \iff \bar{\mathbf{x}} = \mathbf{J}_f(\mathbf{x})^\top \mathbf{v}.$$

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- $\mathbf{v}^\top \mathbf{J}_f(\mathbf{x})$ is a **vector–Jacobian product** (VJP).
- The adjoints propagated in our earlier program are the components of $\bar{\mathbf{x}}$.
- Reverse mode computes this product without constructing the Jacobian itself.

The seed chooses the output combination

Cotangent vector

A cotangent vector is a vector of sensitivities. Applied to a small output change $\delta \mathbf{y}$, it gives the corresponding scalar change $\mathbf{v}^\top \delta \mathbf{y}$.

The output cotangent $\mathbf{v}$ defines a scalar function

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- An arbitrary $\mathbf{v}$ differentiates a weighted sum of the outputs.
- For a scalar output, $m = 1$ and the usual seed is simply $v = 1$.

JVPs and VJPs probe different directions

Forward mode: JVP

$$\mathbf{J}_f(\mathbf{x})\mathbf{u}$$

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Deep learning usually has millions of parameters but a scalar loss: reverse mode gives the complete parameter gradient in one sweep.

Computing a VJP with PyTorch

```
import torch
from torch.func import vjp

def f(x):
    return torch.stack((x[0] * x[1], torch.sin(x[0])))

x = torch.tensor([2., 3.])
y, vjp_fn = vjp(f, x)

v = torch.tensor([1., 0.])
(JTv,) = vjp_fn(v)
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<https://docs.pytorch.org/docs/stable/generated/torch.func.vjp.html>

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- `vjp_fn` maps an output cotangent to input cotangents.
- Here $\text{JT}\mathbf{v}$ is $\mathbf{J}_f(\mathbf{x})^\top \mathbf{v} = [3, 2]^\top$.

<https://docs.pytorch.org/docs/stable/generated/torch.func.vjp.html>

`.backward()` performs the same VJP

```
x = torch.tensor([2., 3.], requires_grad=True)
y = f(x)
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y.backward(v)
print(x.grad)          # tensor([3., 2.])
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Different interface; same mathematics

`torch.func.vjp` returns the VJP values; `.backward()` writes them into the gradients of leaf tensors.

A scalar loss supplies the seed for us

```
optimizer.zero_grad()

prediction = model(inputs)
loss = criterion(prediction, targets)

loss.backward()
optimizer.step()
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https:
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```
loss.backward()
```

```
optimizer.step()
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- Because loss is scalar, `loss.backward()` implicitly uses the seed $v = 1$.

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- The computation graph is freed after the reverse sweep by default.

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Two interfaces, one reverse-mode operation

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- functional: returns values
- makes the cotangent explicit
- composes with other function transforms
- useful for analysis and higher-order methods

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The common idea

Both propagate an output cotangent backwards through the computation graph to evaluate a VJP.